

Ubuntu 运行 AMSS-NCKU-Python 报错信息及处理办法

Debugging AMSS-NCKU-Python on Ubuntu

（不定期更新中）

报错类型一：导入输入文件报错

报错信息

```
import NR_xiaoqu_input
```

报错原因

这是因为用了老的输入文件名 NR_xiaoqu_input.py，没有更新，新的输入文件名为 AMSS_NCKU_Input.py

应对措施

用户可自行将代码更改为：import NR_xiaoqu_input ---> AMSS_NCKU_Input.py

Type of Bugs 1: Errors in importing the input file

Error message

```
import NR_xiaoqu_input
```

Cause of the Error

This is because the old input file name NR_xiaoqu_input.py was used. The new input file is named AMSS_NCKU_Input.py.

Debugging

Users can change the code as: import NR_xiaoqu_input ---> AMSS_NCKU_Input.py

报错类型二：MPI 运行 ABE 报错

Bug Type 2: Errors occurred when running ABE with MPI

报错信息一

Error message 1

There are not enough slots available in the system to satisfy the 64 slots that were requested by the application: ABE

Either request fewer slots for your application, or make more slots available for use.

报错原因

用户电脑没有 64 个 CPU 核心，却使用 MPI 进行 64 核心进行并行计算

Cause of the Error

The user's computer does not have 64 CPU cores, but uses MPI for parallel computing with 64 cores.

应对措施

根据自行电脑的 CPU 核心数设定程序运行时调用的 MPI 进程数

例如，在输入文件 AMSS_NCKU_Input.py 中设定 MPI 并行计算的进程数

MPI_processes = 64 ---> MPI_processes = 8

Debugging

Set the number of MPI processes according to the number of CPU cores in your own computer.

For example, set the number of MPI parallel computing processes in the input file AMSS_NCKU_Input.py.

MPI_processes = 64 ---> MPI_processes = 8

```
1
2 #####
3 ##
4 ## 这个文件包含了数值相对论所需要的输入
5 ## 小曲
6 ## 2024/03/19 --- 2025/12/11
7 ##
8 #####
9
10 import numpy
11
12 #####
13
14 ## 设置程序运行目录和计算资源
15 ## Setting MPI processes and the output file directory
16
17 File_directory = "GW150914_test"          ## 程序运行目录          output file directory
18 Output_directory = "binary_output"        ## 存放二进制数据的子目录    binary data file directory
19                                           ## The file directory name should not be too long
20 MPI_processes = 64                        ## 想要调用的进程数目        number of mpi processes used in the simulation
21
22 GPU_Calculation = "no"                    ## 是否开启 GPU 计算, 可选 yes 或 no    Use GPU or not
23                                           ## (prefer "no" in the current version, because the GPU part may have bugs
24 CPU_Part = 1.0
25 GPU_Part = 0.0
26
27 #####
28
```

报错信息二

未能找到 ABE 生成的文件（包括但不限于 setting.par、Error.log、bssn_BH.dat、bssn_ADMQs.dat、bssn_psi4.dat、bssn_constraint.dat）

No such file or dictionary:

Error message 2

Failed to find the files generated by ABE (including setting.par, Error.log, bssn_BH.dat, bssn_ADMQs.dat, bssn_psi4.dat, bssn_constraint.dat).

No such file or dictionary:

```
Output directory: xiaoqu_Results_GW150914_test/AMSS_NCKU_output
Traceback (most recent call last):
  File "/amssncku-master--bbh/AMSS_NCKU_Program.py", line 351, in <module>
    shutil.copy( AMSS_NCKU_error_file_path, os.path.join(output_directory, "AMSSNCKU_setting_parameter"
  ) )
  File "/usr/lib/python3.10/shutil.py", line 417, in copy
    copyfile(src, dst, follow_symlinks=follow_symlinks)
  File "/usr/lib/python3.10/shutil.py", line 254, in copyfile
    with open(src, 'rb') as fsrc:
FileNotFoundError: [Errno 2] No such file or directory: 'xiaoqu_Results_GW150914_test/AMSS_NCKU_output/out
put_file/setting.par'
```

报错原因

有可能是 ABE（或 ABEGPU）程序运行异常，可进入 AMSS_NCKU_output 文件夹下查看 ABE_out.log（或 ABEGPU_out.log）中的报错信息

有一种报错是如下的行为

Cause of the Error

Maybe the ABE (or ABEGPU) program is running abnormally. You can go to the AMSS_NCKU_output folder and check the error information in ABE_out.log (or ABEGPU_out.log).

One type of error is the following behavior.

```

xiaoqu_Results_GW150914_test > AMSS_NCKU_output > ≡ ABE_out.log
1  -----
2  mpirun has detected an attempt to run as root.
3
4  Running as root is *strongly* discouraged as any mistake (e.g., in
5  defining TMPDIR) or bug can result in catastrophic damage to the OS
6  file system, leaving your system in an unusable state.
7
8  We strongly suggest that you run mpirun as a non-root user.
9
10 You can override this protection by adding the --allow-run-as-root option
11 to the cmd line or by setting two environment variables in the following way:
12 the variable OMPI_ALLOW_RUN_AS_ROOT=1 to indicate the desire to override this
13 protection, and OMPI_ALLOW_RUN_AS_ROOT_CONFIRM=1 to confirm the choice and
14 add one more layer of certainty that you want to do so.
15 We reiterate our advice against doing so - please proceed at your own risk.
16 -----
17

```

表示计算机阻止在 root 用户下运行 mpirun

It indicates that the computer prevents running mpirun under the root user.

应对措施

方案一：在其它（非 root）用户下使用程序

方案二：修改 python 脚本运行 ABE/ABEGPU 程序的命令（`mpirun -np XXX ABE` 或 `mpirun -np XXX ABEGPU`），后面补上 `--allow-run-as-root`

这个命令在 `makefile_and_run.py` 中的 `run_ABE` 函数中

Debugging

Option 1: Use the program under other (non-root) users.

Option 2: Modify the command for the Python script to run the ABE/ABEGPU program (`mpirun -np XXX ABE` or `mpirun -np XXX ABEGPU`), and append `--allow-run-as-root` at the end.

This command is in the function `run_ABE` (in the python file `makefile_and_run.py`).

```

119 #####
120
121 ## 这个函数运行 AMSS-NCKU 主程序 ABE
122
123 def run_ABE( input_language ):
124
125     if ( input_language == "Chinese" ):
126         print(
127             " 正在运行 AMSS-NCKU 主程序 ABE/ABEGPU "
128         )
129     elif ( input_language == "English" ):
130         print(
131             "  Running the AMSS-NCKU executable file ABE/ABEGPU "
132         )
133
134     ## 定义要运行的命令，要使用 str 将其它转换为字符串
135
136     if (input_data.GPU_Calculation == "no"):
137         mpi_command = "mpirun -np " + str(input_data.MPI_processes) + " ./ABE"
138         mpi_command_outfile = "ABE_out.log"
139     elif (input_data.GPU_Calculation == "yes"):
140         mpi_command = "mpirun -np " + str(input_data.MPI_processes) + " ./ABEGPU"
141         mpi_command_outfile = "ABEGPU_out.log"
142
143     ## 使用subprocess.Popen来执行命令，并实时打印输出
144     mpi_process = subprocess.Popen(mpi_command, shell=True, stdout=subprocess.PIPE, stderr=subprocess.STDOUT, text=True)
145
146     ## 将 ABE 的运行结果写入文件中
147     with open(mpi_command_outfile, 'w') as file0:
148         ## 循环读取输出并打印
149         for line in mpi_process.stdout:
150             print(line, end='') # 实时打印输出
151             file0.write(line)    # 将行写入文件
152             file0.flush()       # 确保每行都被立即写入文件，可选
153     file0.close()
154
155     ## 等待进程结束
156     mpi_return_code = mpi_process.wait()
157
158     if ( input_language == "Chinese" ):
159         print(
160             "  AMSS-NCKU 主程序 ABE/ABEGPU 运行结束 "
161         )
162     elif ( input_language == "English" ):
163         print(
164             "  The ABE/ABEGPU simulation is finished "
165         )

```

如有新的报错，请联系我，邮箱 chenkaiqiao@cqut.edu.cn

If there are new error reports, please contact me. Email: chenkaiqiao@cqut.edu.cn